

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
INT332	DEVOPS VIRTUALIZATION AND CONFIGURATION MANAGEMENT	2	0	2	3	
Course Weightage	ATT: 5 CA: 45 ETT: 50					
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT					

Course Outcomes :Through this course students should be able to

CO1 :: Understand the concept of virtualization and containerization.

CO2 :: Understand the implementation and operation of Docker and microservices.

CO3 :: Understand and implement the concept of Docker swarm.

CO4 :: Understand the concept of continuous integration.

CO5 :: Apply the concept of CI/CD using Jenkins.

CO6 :: Develop the software using CI/CD concept.

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	DOCKER DEEP DIVE	NIGEL POULTON	SHROFF/O'REILLY

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	DOCKER: UP AND RUNNING	KARL MATTHIAS, SEAN P. KANE	O'REILLY
R-2	JENKINS CONTINUOUS INTEGRATION COOKBOOK	ALAN MARK BERG	O'REILLY
R-3	JENKINS 2 UP AND RUNNING	BRENT LASTER	O'REILLY

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	4

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Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Introduction to Docker. (Introduction to Docker hub, Docker Architecture, Docker life cycle, Concept of Hyper-V)	T-1		Introduction to Docker.	Understand basics of virtualization and docker.	PPT	NA
	Lecture 2	Introduction to Docker. (Introduction to Docker hub, Docker Architecture, Docker life cycle, Concept of Hyper-V)	T-1		Introduction to Docker.	Understand basics of virtualization and docker.	PPT	NA
Week 2	Lecture 3	Installation of Docker. (Installing Docker on Windows/Linux, Concept of Hyper-V, Basic Docker commands.)	T-1		Installation of Docker.	Learn the installation process of Docker.	PPT	NA
	Lecture 4	Installation of Docker. (Installing Docker on Windows/Linux, Concept of Hyper-V, Basic Docker commands.)	T-1		Installation of Docker.	Learn the installation process of Docker.	PPT	NA
Week 3	Lecture 5	Basic Docker operations. (Create a Docker-hub account, Pushing the Docker to container-hub, Saving changes to a Docker container.)	T-1 R-1		Basic Docker operations.	Learn the usage of various docker commands and docker hub.	PPT	NA
	Lecture 6	Basic Docker operations. (Create a Docker-hub account, Pushing the Docker to container-hub, Saving changes to a Docker container.)	T-1 R-1		Basic Docker operations.	Learn the usage of various docker commands and docker hub.	PPT	NA
Week 4	Lecture 7				BYOD-Practical			
	Lecture 8	Introduction to docker file (Creating a Docker file)	T-1		Introduction to docker file	Creation a docker file and various operations	LIVE DEMONSTRATION	NA

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Week 5	Lecture 9	Introduction to docker file (Creating a Docker file)	T-1		Introduction to docker file	Creation a docker file and various operations	LIVE DEMONSTRATI ON	NA
	Lecture 10	Docker storage(Types of Docker storage, Linking docker container)	T-1		Docker storage.	Understand docker storage.	PPT	NA
Week 6	Lecture 11	Microservices(Introduction to microservices)	T-1		Introduction to microservices	Understand basics of microservices	PPT	NA
	Lecture 12	Microservices(Introduction to microservices)	T-1		Introduction to microservices	Understand basics of microservices	PPT	NA
Week 7	Lecture 13	Microservices(Introduction to microservices)	T-1		Introduction to microservices	Understand basics of microservices	PPT	NA
SPILL OVER								
Week 7	Lecture 14				Spill Over			
MID-TERM								
Week 8	Lecture 15	Container orchestration (Introduction to container orchestration)	T-1		Container orchestration	Introduction to container orchestration	PPT	NA
	Lecture 16	Docker swarm cluster (Creating a Docker swarm cluster, Initializing Docker Swarm create a service in a Docker Swarm)	T-1		Docker swarm cluster	Learn to create docker swarm cluster	LIVE DEMONSTRATI ON	NA
Week 9	Lecture 17	Docker swarm cluster (Creating a Docker swarm cluster, Initializing Docker Swarm create a service in a Docker Swarm)	T-1		Docker swarm cluster	Learn to create docker swarm cluster	LIVE DEMONSTRATI ON	NA
	Lecture 18	Docker networks (Introduction to Docker networks, Creating a service)	T-1		Introduction to Docker networks, Creating a service	Understand the basics of docker networks	LIVE DEMONSTARTI ON	NA
Week 10	Lecture 19	Continuous integration (Introduction to continuous integration.)	R-2		Continuous integration	Understand the concept of continuous integration	PPT	NA
	Lecture 20	Continuous integration (Introduction to continuous integration.)	R-2		Continuous integration	Understand the concept of continuous integration	PPT	NA
Week 11	Lecture 21				Project			

Week 11	Lecture 22	Jenkins(Jenkins Architecture, Installing Jenkins)	R-2 R-3		Jenkins	Understand Jenkins architecture and installation of Jenkins	LIVE DEMONSTRATION	
Week 12	Lecture 23	Jenkins(Jenkins Architecture, Installing Jenkins)	R-2 R-3		Jenkins	Understand Jenkins architecture and installation of Jenkins	LIVE DEMONSTRATION	
	Lecture 24	Node Management (Managing nodes on Jenkins)	R-2		Jenkins Integration	Learn Jenkins integration with devops tools	LIVE DEMONSTRATION	NA
Week 13	Lecture 25	Node Management (Managing nodes on Jenkins)	R-2		Jenkins Integration	Learn Jenkins integration with devops tools	LIVE DEMONSTRATION	NA
	Lecture 26	CI / CD Pipelines (Understanding CI / CD Pipelines, Creating CI / CD Pipelines)	R-2		CI / CD Pipelines	Understanding CI / CD Pipelines, Creating CI / CD Pipelines	LIVE DEMONSTRATION	NA
Week 14	Lecture 27	CI / CD Pipelines (Understanding CI / CD Pipelines, Creating CI / CD Pipelines)	R-2		CI / CD Pipelines	Understanding CI / CD Pipelines, Creating CI / CD Pipelines	LIVE DEMONSTRATION	NA
		SPILL OVER						
Week 14	Lecture 28				Spill Over			
Week 15	Lecture 29				Spill Over			
	Lecture 30				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0202 (2 out of 2)

Component	Weightage (%)	Mapped CO(s)
Project	50	CO1, CO2, CO3, CO4, CO5, CO6
BYOD-Practical	50	CO1, CO2, CO3

Details of Academic Task(s)

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Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
BYOD-Practical	To evaluate progress of individual student based on Docker.	BYOD Practical	Individual	Online	30	4 / 5
Project	Practical application of the concepts learnt regarding docker and Jenkins.	Project	Individual	Online	100	4 / 12

MOOCs/ Certification etc. mapped with the Academic Task(s)

Academic Task	Name Of Certification/Online Course/Test/Competition mapped	Type	Offered By Organisation
BYOD-Practical	AWS CERTIFIED DEVOPS ENGINEER - PROFESSIONAL	Industry Certification	AMAZON WEB SERVICES

- Where MOOCs/ Certification etc. are mapped with Academic Tasks:
1. Students have choice to appear for Academic Task or MOOCs etc.
 2. The student may appear for both, In this case best obtained marks will be considered.

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	List of practical.	Installing Docker on Windows/Linux, Concept of Hyper-V, Basic Docker commands.		Working of docker and Jenkins
Practical 2	List of practical.	Create a Docker-hub account, Pushing the Docker to container-hub, Saving changes to a Docker container.		Learn the usage of various docker commands and docker hub.
Practical 3	List of practical.	Creating a Docker file		Creation a docker file and various operations
Practical 4	List of practical.	Types of Docker storage, Linking docker container		Understand docker storage
Practical 5	List of practical.	Introduction to micro services		Understanding of Microservices
Practical 6	List of practical.	Docker compose installation, Introduction to YAML files, Writing a Docker Compose file, deploying WordPress using docker compose.		Introduction to docker compose
Practical 7	List of practical.	Introduction to container orchestration		Container orchestration
Practical 8	List of practical.	Creating a Docker swarm cluster, Initializing Docker Swarm create a service in a Docker Swarm.		Learn to create docker swarm cluster

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Practical 9	List of practical.	Introduction to Docker networks, Creating a service.		Understand the basics of docker networks
Practical 10	List of practical.	Introduction to continuous integration.		Understand the concept of continuous integration
Practical 11	List of practical.	Jenkins Architecture, Installing Jenkins.		Understand installation of Jenkins.
Practical 12	List of practical.	Managing nodes on Jenkins.		Learn node management on Jenkins
Practical 13	List of practical.	Understanding CI / CD Pipelines, Creating CI / CD Pipelines.		Creating CI / CD Pipelines
	SPILL OVER			
Practical 14	Spill Over			